

Aerodynamic Prediction Methodology for Grid Fins

Richard W. Kretschmar

Missile Research Development and Engineering Center
U.S. Army Aviation and Missile Command
Attn: AMSAM-RD-SS-AT
Redstone Arsenal, AL 35898-5000, USA

Dr. John E. Burkhalter

Department of Aerospace Engineering
Auburn University, AL 36849, USA

SUMMARY

Formulation and validation of a theoretical methodology to predict the aerodynamic forces and moments associated with missile configurations utilizing grid fins in subsonic, transonic and supersonic Mach regimes is discussed in detail. Comparisons with experimental data collected on a variety of grid fin configurations are presented to provide an indication of the methodology accuracy. Conclusions regarding the accuracy and limitations of the prediction methodology are drawn from these comparisons.

1.0 LIST OF SYMBOLS

A	Subelement Area
B_z	Fin Span from Body Centerline
B_t	Fin Span
C	Fin Chord Length
C_F	Element Force Coefficient
C_N	Element Normal Force Coefficient
C_P	Element Side Force Coefficient
C_A	Element Axial Force Coefficient
H	Fin Height
H_b	Height of Fin Support Structure
i_{base}	Number of Cells in Fin Base Corner
i_{tip}	Number of Cells in Fin Tip Corner
M	Mach Number
n_{dy}	Number of Cells in Horizontal Direction
n_{dz}	Number of Cells in Vertical Direction
$\vec{n}$	Component of Unit Normal Vector
S	Element Slant Length
S_{ref}	Reference Area
$\vec{u}$	Component of Velocity
y_{fb}	Span of Fin Support Structure
X_h	Longitudinal Position of Fin Hinge Line
α	Angle of Attack
β	Compressibility Factor
δ	Fin Deflection Angle
Γ	Vortex Strength
ϕ	Fin Azimuthal Orientation Angle
θ	Azimuthal Angle from Horizontal Y axis

2.0 INTRODUCTION

The Missile Research Development and Engineering Center of the U.S. Army Aviation and Missile Command has developed a prediction methodology to accurately predict the aerodynamic forces and moments associated with missile configurations utilizing grid fins as lift and control surfaces.

Flow field analyses conducted by Russian researchers have provided a basic understanding of and highlighted the substantial differences in flow field characteristics, particularly the shock structure, resulting from the presence of grid fins in various Mach regimes.¹ These differences required separate development of theoretical formulations for the subsonic, transonic and supersonic Mach regimes.

Independent of the Mach regime, the unique grid fin geometry requires prediction techniques which address multiple lifting surfaces oriented at angles other than the traditional "wing" plane. Existing theoretical techniques, such as vortex lattice theory for subsonic applications and Evvard's theory for supersonic applications, lend themselves to this type of analysis and were therefore applied directly or in slightly modified form to the grid fin problem.

Due to the linear nature of the selected theoretical formulations, their application was limited to missile orientations at which nonlinear aerodynamic characteristics are minimal i.e. small angles of attack. Investigation of experimental data has indicated linear aerodynamic characteristics are maintained up to approximately 7.0 deg. for most grid fin configurations. To extend the applicable range of these theories, a series of empirical equations were developed to correlate the prediction results for both subsonic and supersonic cases with experimental data collected in a series of wind tunnel tests. Limited validation of the theoretical formulations has been accomplished through comparisons with experimental data.

3.0 GRID FIN GEOMETRIC DEFINITION

Theoretical formulations developed in this analysis are quite robust and apply to nearly all grid fin shapes of practical interest. However, some geometric restrictions are necessary to limit the programming complexity related to the application of these theories. Currently, application is limited to grid fin configurations which are symmetric about the "Y" axis and have a hinge line which is coincident with a radial vector from the body centerline outward. Additionally, the geometry of each fin within a fin set must be identical.

Due to the complexity of the grid fin geometry, numerous geometric parameters are required for accurate definition. In addition to the obvious parameters such as fin span, height, chord length and longitudinal position on the missile body, parameters to define the span and height of the support structure, the number of cells in the vertical and horizontal directions and the number of cells in the inboard and outboard corners are also required. An illustration of these parameters is provided in Figure 1.

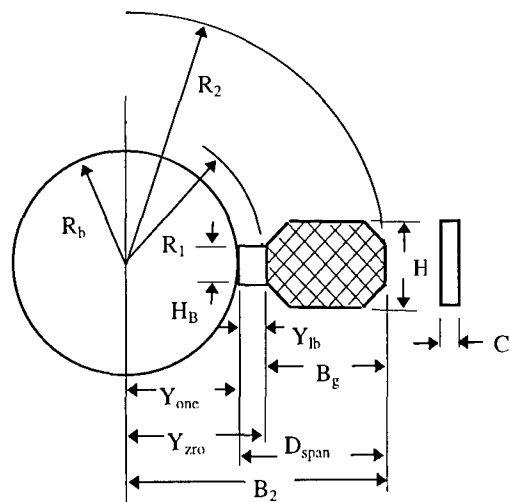


Figure 1. Geometric Parameters Used to Define Grid Fin Geometry

Two angles, the azimuthal and deflection angle must be specified to define the orientation of each individual fin relative to the missile body. Fin azimuthal position is defined in the Y-Z plane and is positive counterclockwise from the Y axis. Fin deflection angle is defined such that a positive fin deflection results in a counterclockwise roll of the missile and is referenced about the fin hinge line. Sign convention for each of these angles is illustrated in Figure 2.

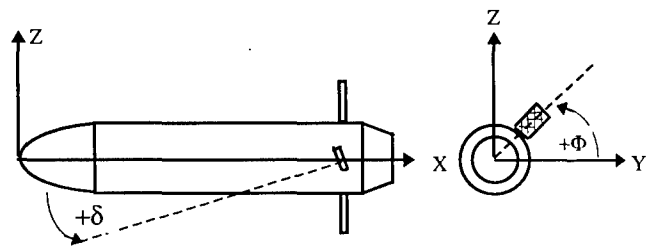


Figure 2. Orientation Angle Sign Convention.

4.0 THEORETICAL FORMULATION

Selection of the proper theoretical formulation is governed by the freestream Mach number being considered. Analysis of existing wind tunnel data has provided a general understanding of the flow field surrounding a generic grid fin design.^{2,3} Table 1 provides a descriptive summary of the assumed flow structure and specifies the applicable theoretical formulation.

Table 1. Summary of Assumed Flowfield Structure.

Mach Number	Formulation	Assumed Flow Structure
$M < 0.8$	Subsonic	Local Flow structure completely subsonic. Compressible subsonic formulation is required.
$0.8 < M < 1.0$	Transonic	Flow through Grid Structure is Typically Choked Requiring Application of Coefficient Correction Factors.
$1.0 < M < 1.4$	Transonic	Bow shock in front of grid fin. Flow is subsonic behind the bow shock and a compressible subsonic solution is required.
$1.4 < M < 1.9$	Supersonic	Grid fin swallows shock. Each element acts as a thin wing with an attached leading shock which impinges on adjacent fin elements producing a reflected shock dominated flow region.
$1.9 < M < 3.5$	Supersonic	Leading edge shock on each fin element does not impinge on adjacent elements and internal flow is primarily supersonic with minimal reflected shock effects
$M > 3.5$	Supersonic	Strong leading edge shock transitioning to hypersonic flow.

4.1 Subsonic Formulation

The theoretical formulation developed for the subsonic Mach regime utilizes a vortex lattice solution to calculate the loading on each individual element in the lattice.⁴ Body upwash and wing-body carryover terms resulting from interaction between the missile components are also accounted for using theoretical techniques.

Fin geometry is modeled by replacing the grid fin with a vortex lattice structure. This lattice consists of a series of horseshoe vortices and associated control points and normal vectors affixed to each individual panel element. Defining the vortex lattice structure is a multi-stpped process outlined in detail in Reference 4 and summarized here for clarity. The process begins by defining the intersection points between the panels making up the grid fin structure. When these points are connected and given a chord depth, the three dimensional aspects of the grid fin are established. If one considers a single element of a grid fin located arbitrarily in the freestream, the element would appear as a flat plate as illustrated in Figure 3.

A horseshoe vortex is affixed to each panel and is defined using 10 node points. Node points 1 and 6, located at the panel quarter chord and node points 2 and 7, located at the trailing edge of the panel define the bound portion of the vortex. The remaining 6 nodes define the vortex trailing legs and are positioned to trail off in the direction of the freestream. A control point and unit normal vector are positioned at the midspan and three quarter chord of each panel, respectively.

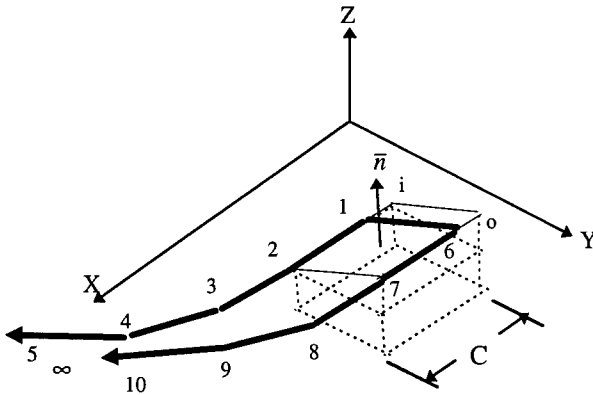


Figure 3. Schematic of an Individual Grid Fin Element and Associated Vortex Lattice Structure.

To determine the loading on each grid fin, the strengths of each vortex must first be determined. This is accomplished by applying boundary conditions requiring the flow to be tangent to each panel at the control point. The velocity vector at a control point is composed of three components, the freestream velocity, the crossflow velocity or body upwash component, and the induced velocity from each vortex filament in the flowfield. Determination of the cross flow velocity magnitude is accomplished using a potential flow solution for an infinite doublet in a freestream. An illustration of the flow field generated by a doublet in a freestream is provided in Figure 4.

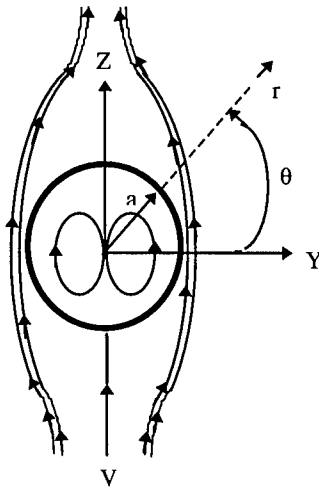


Figure 4. Flowfield Resulting from a Freestream Doublet Combination.

General expressions for the velocity components in Cartesian coordinates resulting from the potential flow solution of an infinite doublet are,

$$u_y = -\sin \alpha \sin 2\theta \left(\frac{r_b^2}{|r|^2} \right)$$

$$u_z = \sin \alpha \left[1 + (1 - 2 \sin^2 \theta) \left(\frac{r_b^2}{|r|^2} \right) \right]$$

The compressible form of the Biot-Savart integral is used to determine the velocity induced by a vortex filament segment at a control point and may be written as,

$$\bar{v}_r = -\frac{\Gamma \beta^2}{4\pi} \int \frac{\bar{r} x d\bar{l}}{|r_\beta|^3}$$

where β is the compressibility factor,

$$\beta = \sqrt{1 - M^2}$$

The velocity vector at the control point can be determined by summing the contributions of the freestream doublet combination and the velocities induced by the vortex filaments.

$$\bar{v}_i = \bar{v}_r + u_x \hat{i} + u_y \hat{j} + u_z \hat{k}$$

The dot product of the velocity vector and the panel unit normal yields the component of velocity normal to the panel surface. For the flow tangency boundary condition to be satisfied, this normal velocity must be zero. Application of this condition yields an equation which when reorganized isolates the unknown vortex strengths and is conducive to solution using numerical methods

$$\bar{v}_r \cdot \bar{n} = \left[\left(-\frac{\beta^2}{4\pi} \int \frac{\bar{r} x d\bar{l}}{|r_\beta|^3} \right)_x \cdot n_x + \left(-\frac{\beta^2}{4\pi} \int \frac{\bar{r} x d\bar{l}}{|r_\beta|^3} \right)_y \cdot n_y + \left(-\frac{\beta^2}{4\pi} \int \frac{\bar{r} x d\bar{l}}{|r_\beta|^3} \right)_z \cdot n_z \right] \cdot \Gamma = B$$

Application of the above equation to all vortices in the flowfield yields an expression which is best expressed in matrix format.

$$[\Gamma] = [A]^{-1} \cdot [B]$$

In the above expression, the "A" matrix is the inverse of the bracketed term above, the "B" matrix represents the known velocities induced at the control point by the freestream doublet combination, and the "Γ" term is the matrix of unknown vortex strengths.

Vortex strengths are determined simultaneously using an iterative procedure to avoid problems in inverting the often large "A" matrix. The solution process for a missile with 4 fins is as follows; First, vortex strengths of fin #1 are found as if no other fins were present. Second, the vortex strengths of fin #2 are determined as if only fin #1 and #2 exist and the strengths of the vortices on fin #1 are known. This process continues for the remaining fins, each time including the known vortex strengths of the previous grid fins. When the vortex strengths of fin #4 have been found, the entire process is repeated for several iterations until the updated vortex strengths are no longer significantly different from the values calculated in the previous iteration. In this manner, inversion of large matrices is avoided. From past experience, only a few iterations are necessary to obtain a converged solution.

Aerodynamic loads on the grid fin can be computed directly from the calculated vortex strengths by considering the loading on each individual panel separately. Application of

the Kutta-Joukowski theorem given below yields the force coefficient normal to the panel surface where S is the slant length of the element.

$$C_{\hat{F}} = \frac{2 \cdot \Gamma \cdot S}{S_{ref}}$$

Components of this force in the master coordinate system are then determined by multiplication with the unit normal vector.

$$C_{\hat{N}} = C_{\hat{F}} \cdot \bar{n}_z$$

$$C_{\hat{Y}} = C_{\hat{F}} \cdot \bar{n}_y$$

$$C_{\hat{A}} = C_{\hat{F}} \cdot \bar{n}_x$$

Fin-body carryover loading is accounted for using an imaging scheme in which each panel element is imaged inside the body in a particular geometric fashion as illustrated in Figure 5.

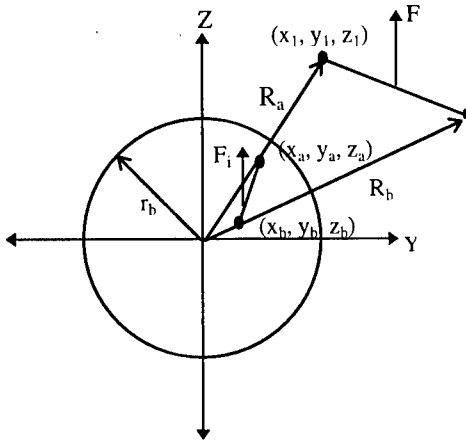


Figure 5. Illustration of Imaging Scheme Used in Fin-Body Carryover Calculation

Orientation of the imaged panel is governed by the position of the actual panel. Endpoints of the imaged panel are positioned along a radial line from the endpoints of the actual panel to the body centerline. The governing assumption for the imaged panel is that the normal force per unit span for the actual panel element and the imaged panel element is the same. This is expressed in coefficient form as

$$C_{Ni} = C_N \cdot \left(\frac{S_{abi}}{S_{ab}} \right)$$

$$C_{Si} = C_S \cdot \left(\frac{S_{abi}}{S_{ab}} \right)$$

where, S_{abi} and S_{ab} are the spanwise lengths of the imaged and actual panels, respectively.

Grid fin axial force is assumed to be a composition of induced drag, skin friction, pressure drag and interference drag from the fin element intersections. Induced drag is defined as the component of the fin force vector that is projected into the axial direction (parallel to the body) due to fin incidence.

$$C_{Ai} = C_N \tan \delta$$

For skin friction drag, the fin wetted area is calculated and converted to a flat plate area with an assumed laminar or turbulent boundary layer. A skin friction coefficient is then determined as a function of Reynolds number, from which a skin friction axial force coefficient (C_{axf}) is determined. These computations were derived from those presented in Reference 6. Pressure drag (C_{adp}) is considered a product of the local dynamic pressure and the fin frontal area and is written in coefficient form as

$$C_{Adp} = \frac{S_{wet} t}{2 \cdot S_{ref} C}$$

Interference drag resulting from the fin element intersections was determined empirically from experimental data⁷ as

$$C_{Axp} = 2 \cdot 0.000547 \cdot (np + 2)$$

where $(np + 2)$ is the total number of grid intersection points including points on the base support structure. Total fin axial force is then determined by summing the above components.

$$C_{Ax} = C_{Axi} + C_{Axf} + C_{Aadp} + C_{Axp}$$

Observations of experimental data have indicated that fin axial force changes very little with angle of attack.^{2,8} It is therefore assumed independent of angle of attack and given a value based on the above equation for some given value of fin incidence angle.

Nonlinear Aerodynamic Formulation

Because the grid fin element is in effect a very "thin" wing, it begins to lose lift at angles of attack around 7.0 deg, resulting in nonlinear characteristics at higher angles of attack. This characteristic precludes the sole use of a linear theory such as vortex lattice theory in predicting the grid fin aerodynamics. However, vortex lattice theory does provide an accurate estimate of the initial lift curve slopes forming a basis from which a nonlinear theory can be developed.

For grid fins, accurate extension of prediction capability to higher angles of attack using an empirical method is strongly dependent on an accurate computation of grid fin normal force. Observations of wind tunnel data for several different grid fin configurations indicate that grid fins do not stall in the traditional sense, but continue to produce lift at very high angles of attack.⁹ This observation drove the development of an equation, provide below, which seems to fit the available experimental normal force curves.

$$C_N = \left\{ \frac{\frac{C_{N\alpha} \sin \alpha}{1 + k_1 \left(\frac{B_g}{H} \right)^2 \left(\frac{C}{H} \right) \sin^2 \alpha} + \frac{\frac{C_{N\delta} \sin \delta}{1 + k_2 \left(\frac{B_g}{H} \right)^2 \left(\frac{C}{H} \right) \sin^2 \delta}}{\left(\frac{k_3 - k_4 \sqrt{\frac{B_g}{H}}}{k_6 - \frac{\left(\frac{B_g}{H} \right)^2}{1 + k_5 \left(\frac{B_g}{H} \right)^2} \alpha^2} C_{N\alpha} \sin \alpha \right)} \right\} \cos^2(2\alpha - \delta)$$

This semiempirical formulation utilizes the initial lift curve slopes from vortex lattice theory and attempts to incorporate the influence of the most prominent geometric parameters effecting the grid fin aerodynamic characteristics, the ratio of fin span to height (b_g/h) and the ratio of fin chord length to height (C_h/h). A complete summary of the nonlinear formulation development can be found in reference 5.

4.2 Transonic Aerodynamics

Unlike conventional planar fins which experience maximum normal force coefficient values at transonic Mach numbers, a grid fin experiences what is commonly referred to as the "transonic bucket". To understand this phenomenon, one must look at a grid fin as a collection of individual cells acting as separate inlets.⁹

Reduction in the inlet cross-sectional area caused by the presence of the cell walls and the boundary layer buildup on these walls causes the flow passing through the cell to accelerate to sonic conditions (i.e., become choked) at freestream Mach numbers less than 1.0. As the flow increases past Mach 1.0, a detached normal shock forms in front of the cell. The cell remains choked until the normal shock is swallowed. It is the flow spillage caused by the choked flow that causes the reduction in normal force. An illustration of this transonic bucket is presented in Figure 6.⁹

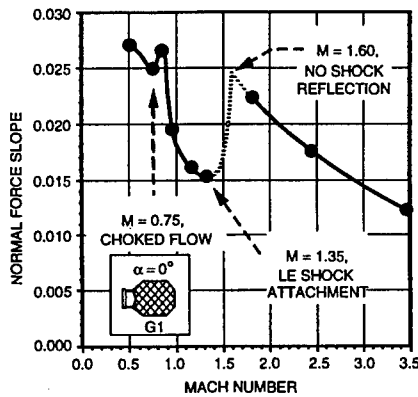


Figure 6. Reduction of Grid Fin Normal Force in Transonic regime.

To accommodate the transonic flow characteristics at Mach numbers less than 1.0, a theoretical formulation was developed to determine whether choked conditions exist for the particular subsonic Mach number / grid fin combination being investigated. In addressing this phenomenon, a laminar boundary layer and isentropic conditions are assumed within the cell structure. Blasius' theorem is utilized to calculate the displacement thickness associated with the local Reynolds number and chord length.

$$\delta_{th} = \frac{1.7208 * C}{\sqrt{Re_c}}$$

By assuming this displacement thickness is present on all fin elements, the corresponding reduction in fin capture area can be determined.

$$A_{\delta_{th}} = 2 * \delta_{th} + thk * ttle$$

Where "ttle" is the total length of the grid fin elements in the plane perpendicular to the freestream. By subtracting this area from the total capture area of the fin, the exit area A_{ex} is determined.

For a given freestream Mach number, there exists a minimum exit area known as the throat area (A^*) at which sonic flow will occur. The ratio between this area and the flow capture area is determined using the isentropic relationship.

$$A^* = \frac{A_{cap} * M}{\left[\frac{2 \left(1 + \frac{\gamma-1}{2} M^2 \right)}{\gamma+1} \right]^{\frac{1}{2} \left(\frac{\gamma+1}{\gamma-1} \right)}}$$

If the calculated exit area A_{ex} is less than A^* the flow is considered choked and a correction factor is determined by calculating the reduction in mass flow rate between the choked and unchoked conditions. The correction factor is determined using the following equation.

$$CF = \frac{A_{ex} * \sqrt{\gamma P_o \rho_o \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}}}}{A_{cap} \rho_{\infty} V_{\infty}}$$

By utilizing this correction factor as a multiplier to reduce the calculated fin forces and moments, the loss of fin effectiveness due to the reduction in mass flow through the grid fin is quantified.

For transonic Mach numbers above Mach 1.0, it is assumed a normal shock exists in front of the entire grid fin surface. After passing through the normal shock wave, the flowfield experienced by the grid fin is again subsonic. Mach number and dynamic pressure behind the shock are calculated using historical normal shock relations and are then applied in the subsonic prediction formulation. Equations relating static pressure and Mach number in front of and behind a normal shock are presented below.

$$\frac{p_2}{p_1} = \frac{2\gamma M_1^2 - (\gamma - 1)}{\gamma + 1}$$

$$M_2^2 = \frac{(\gamma - 1) M_1^2 + 2}{2\gamma M_1^2 - (\gamma - 1)}$$

An illustration of the transonic flowfield structure surrounding a typical grid fin is provided in Figure 7.⁹

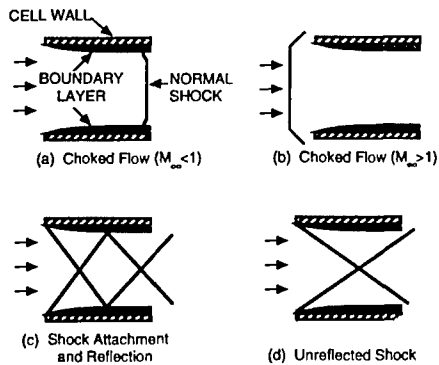


Figure 7. Flowfield Structure Around a Grid Fin at Transonic Conditions.

4.3 Supersonic Aerodynamics

Prediction formulation developed for supersonic flow is similar to the subsonic formulation in that an existing linear aerodynamic technique is applied to the problem by considering the grid fin on a panel by panel basis. In this case, a modified form of Evvard's theory was utilized in the linear aerodynamic range. Modifications to the basic theory were necessary to account for the endplate effects resulting from the grid structure when addressing the individual panel elements. In its original form, Evvard's theory does not properly address this condition.

The original form of Evvard's theory provides generalized formulas for ΔC_p distributions for the case of a swept wing planform with a supersonic leading edge. Figure 7 illustrates the various supersonic regions associated with a typical planar wing. Figures 8 and 9 indicate the regions associated with a grid fin element taking into account the endplate effects for cases in which the Mach lines emanating from the intersection points do not intersect one another and do intersect one another, respectively. As indicated, for a grid fin element, pressure calculations for regions 1, 2 and perhaps 4 are of practical interest.

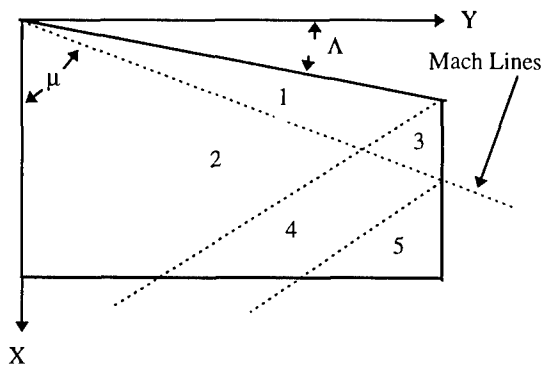


Figure 7. Regions of Influence on a Planar Wing in Supersonic Flow.

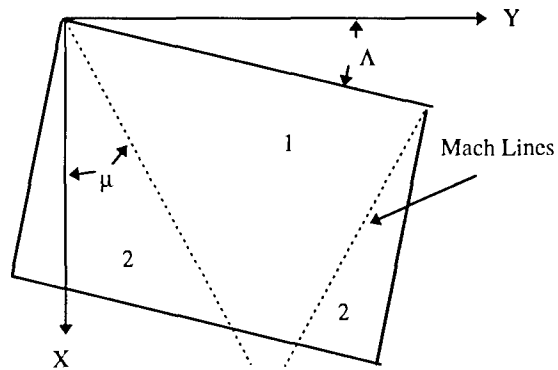


Figure 8. Regions of Influence on a Grid Fin Element in Supersonic Flow, Without Mach Line Interaction.

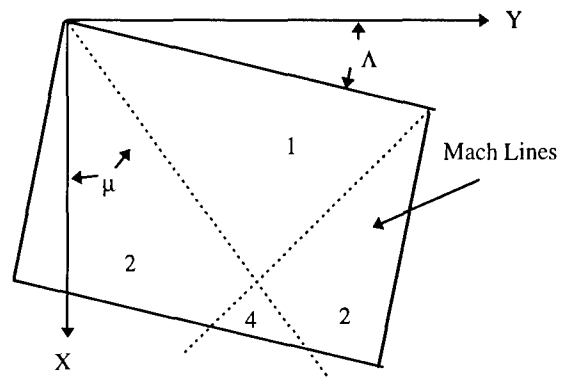


Figure 9. Regions of Influence on a Grid Fin Element in Supersonic Flow, with Mach Line Interaction

Equations which determine the differential pressure loading in regions 1, region 2 and region 4 were developed by Evvard and are presented below.

Region 1:
$$\Delta C_p = \frac{4\alpha}{\sqrt{M^2 - 1 - \tan^2 \lambda}}$$

Region 2:

$$\Delta C_p = \frac{4\alpha}{\pi \sqrt{B^2 - \tan^2 \lambda}} \left[\cos^{-1} \left(\frac{x \tan \lambda - B^2 y}{B(x + y \tan \lambda)} \right) + \cos^{-1} \left(\frac{x \tan \lambda - B^2 y}{B(x - y \tan \lambda)} \right) \right]$$

Region 4:

$$\Delta C_p = \left[\Delta C_p \right]_{reg 2} - \frac{4\alpha}{\pi \sqrt{B^2 - \tan^2 \lambda}} \left[-\cos^{-1} \left(\frac{x_a + y_a (2B + \tan \lambda)}{(x_a + y_a \tan \lambda)} \right) \right]$$

To compute the loading on each element, the element is subdivided into a series of small rectangles. On each of these subelements, a ΔC_p value, determined using the appropriate equations for the particular region in which the subelement resides, is applied to the subelement area A . Equations for the various forces are presented below.

$$\Delta C_N = \frac{\Delta C_p AU_{NZ}}{S_{ref}}$$

$$\Delta C_A = \frac{\Delta C_p AU_{NX}}{S_{ref}}$$

$$\Delta C_Y = \frac{\Delta C_p AU_{NY}}{S_{ref}}$$

Incremental moment coefficients are determined as follows,

$$\Delta C_{lm} = -\Delta C_N \frac{x_{dis}}{L_{ref}} + \Delta C_A \frac{z_{dis}}{L_{ref}}$$

$$\Delta C_{bm} = -\Delta C_N \frac{y_{dis}}{L_{ref}} + \Delta C_Y \frac{z_{dis}}{L_{ref}}$$

$$\Delta C_{ym} = -\Delta C_Y \frac{x_{dis}}{L_{ref}} + \Delta C_A \frac{y_{dis}}{L_{ref}}$$

Force and moment coefficients for the entire fin are determined by summing the force contributions from the individual subelements.

Fin-body carryover effects are accounted for through application of the same imaging scheme described in the subsonic formulation summarized in section 5.1. Similarly, in determining the total fin axial force, the component contribution summation described for the subsonic theoretical formulation in section 5.1 is also included in the supersonic theoretical formulation.

Nonlinear Aerodynamic Formulation

As in the subsonic case, observations of available experimental data indicate significant nonlinear aerodynamic characteristics in the supersonic Mach regime at angles of attack above approximately 7.0 deg. In this regime, the nonlinearities are attributed to the complex shock structure associated with the grid fin configurations. These nonlinearities again preclude the sole use of a purely linear theory such as Evvard's solution in predicting the aerodynamic characteristics of grid fins. However, similar to the subsonic formulation, Evvard's theory is used to determine the initial lift curve slopes of the grid fin configurations, forming a basis for the nonlinear theory.

Unlike the subsonic formulation, only minimal experimental data were available with which to develop the supersonic nonlinear formulation. In this case, the parametric equation which seems to fit the prediction of grid fin normal force may be written as

$$C_N = \left. \frac{C_{N\delta} \delta}{1 + \frac{\delta}{\delta_{max}}} \right|_{\alpha=0} + \left. \frac{C_{N\alpha} \alpha}{1 + \frac{\alpha}{\alpha_{max}}} \right|_{\delta=0} * \left(1 - \left. \frac{C_{N\delta} \delta}{1 + \left(\frac{\delta}{\delta_{max}} \right)^2} \right|_{\alpha=0} \right)$$

This equation has the required linear limit as either α or δ approach zero and reaches a maximum value at α_{max} or at δ_{max} . A complete summary of the development of this formulation can be found in reference 11.

5.0 COMPARISONS WITH EXPERIMENTAL RESULTS

Results from the theoretical formulations have been compared with wind tunnel test data for various fin alone configurations to illustrate the accuracy and identify the limitations of the prediction techniques. The primary concentration in this validation effort has been determining the accuracy of the theoretical formulations in predicting the normal force and axial force of various grid fin configurations in the horizontal position at fin deflection angles of zero deg. This concentration is necessary to limit the scope of the paper and is justified in that accurate grid fin force prediction is the basis for all other aerodynamic characteristic estimation. Additionally, although the nonlinear formulations include the fin deflection parameter " δ ", due to the lack of sufficient wind tunnel test data, no effort was made to validate the contribution of this parameter to the overall aerodynamic characteristics of the grid fin configurations.

Wind Tunnel Model Configurations

A schematic of the grid fin Missile body combination with an arbitrary set of fins is provided in Figure 10. As indicated, the test article consisted of a 5.0 in. body of revolution 52. in. in length, with a three caliber tangent ogive nose. Four grid fins were mounted in the cruciform orientation with grid fins 2 and 4 in the horizontal and grid fins 1 and 3 in the vertical positions.

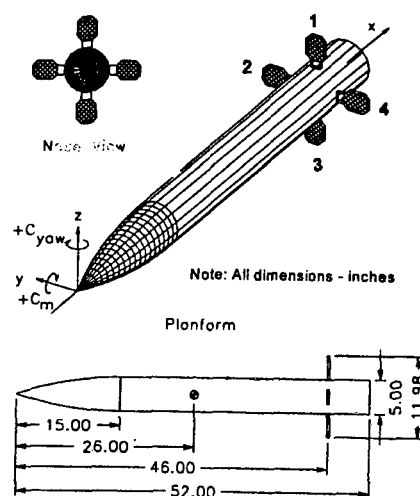


Figure 10. Grid Fin Missile Configuration.

Grid fin configurations utilized in this comparison represent a parametric set of configurations which address the majority of geometric parameters effecting their aerodynamic characteristics. Geometric parameters addressed include, fin span, cell density and fin chord length. Variation of these specific parameters are made on a fin by fin basis while holding the remaining geometric parameters constant. The effect of fin span variations are addressed by fins “G12” and “G13”. Cell density variations are addressed using fins “G12” and “G14” as well as fins “G13” and “G15”. Fin chord variation effects are investigated using fins “G15” and “G16”. Regardless of fin geometry, the fin base structure is consistent for all configurations tested. Illustrations of the various grid fin configurations are provided along with their designations in Figure 11.

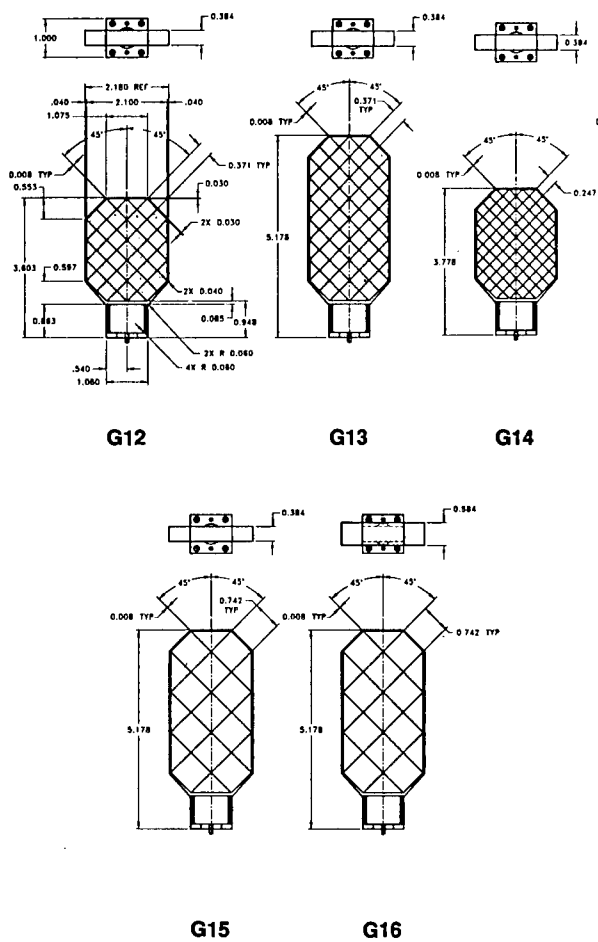


Figure 11. Grid Fin Configurations Used in Validation Process.

Experimental Accuracy

Experimental data presented in this report was collected using a 6-component fin balance mounted inside the wind tunnel body. Quoted accuracy for the fin balances is 1.5% of the rated loads, which results in an accuracy of ± 2 lb. for normal force. Repeatability proved much better, quoted to be within 0.1% of the rated loads. In all experimental test data, the reference length and area are the missile base diameter (5.0 in.) and base area (19.63 in²), respectively.

Analysis

Figures 12 through 14 illustrate the comparison between wind tunnel data and theoretical computations of normal force and axial force coefficient for a single “G13” fin at Mach 0.25, Mach 0.5 and Mach 0.7, respectively. As indicated, the theoretical computations provide an excellent approximation of the initial normal force curve slope at all three subsonic Mach numbers. As a result, the agreement between the theoretical and experimental data is excellent through the linear aerodynamic range to approximately 7 deg. angle of attack. At higher angles of attack, the semiempirical formulation used to predict the nonlinear portions of the curve correctly approximates the curve trend, but significantly underpredicts the normal force coefficient magnitude. Axial force coefficient is slightly overpredicted at Mach 0.25 at small angles of attack. However, at Mach 0.5 and 0.7, CA is accurately predicted throughout the angle of attack range.

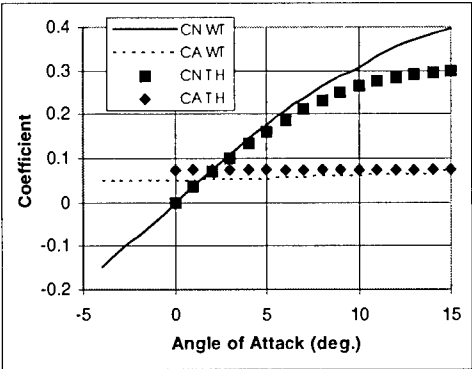


Figure 12. CN and CA Comparison for Fin G13 at Mach 0.25.

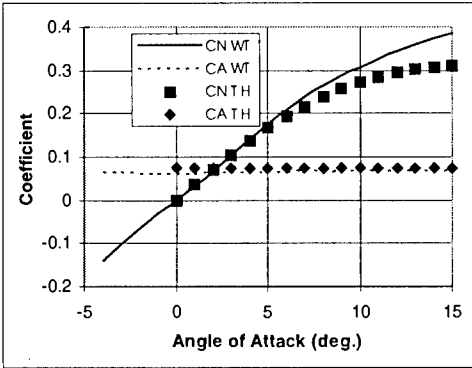


Figure 13. CN and CA Comparison for Fin G13 at Mach 0.5.

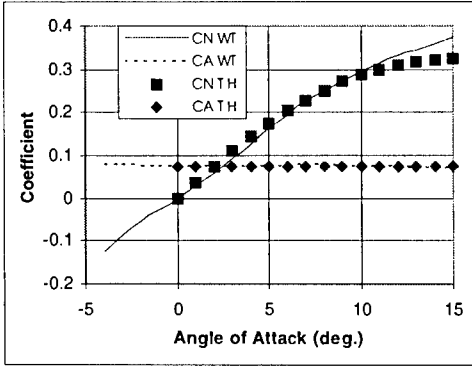


Figure 14. CN and CA Comparison for Fin G13 at Mach 0.7

Significant improvement in the normal force coefficient prediction is evident at both Mach 0.5 and 0.7 as illustrated in Figures 15 and 16, respectively. As indicated, the initial coefficient slope is accurately predicted and the nonlinear aspect of the curves are captured throughout the angle of attack ranges. Axial force coefficient is also predicted accurately for this configuration although a slight overprediction is evident for this coefficient at Mach 0.7.

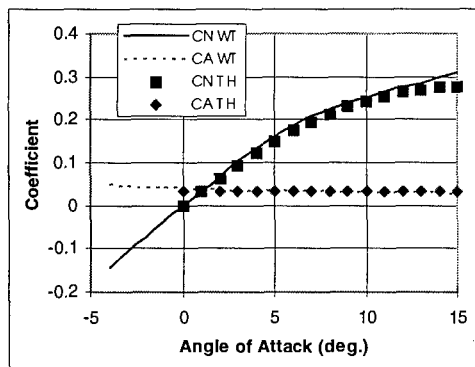


Figure 15. CN and CA Comparisons for Fin G15 at Mach 0.5.

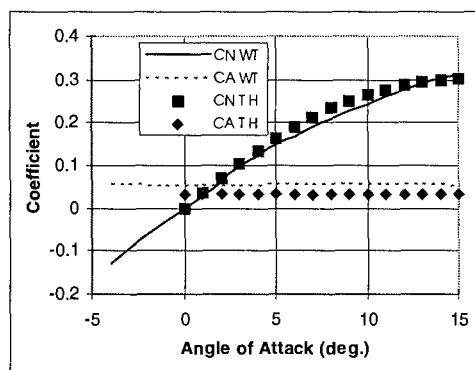


Figure 16. CN and CA Comparisons for Fin G15 at Mach 0.7.

A slight shortcoming in the theoretical formulations prediction accuracy is exhibited in Figures 17 and 18, which illustrate the CN and CA comparisons for fin G16 at Mach 0.5 and 0.7, respectively. As indicated in Figure 11, the geometric difference between fin G16 and G15 is the 0.2 in. increase in fin chord length for G16. This additional lifting area provides a significant increase in normal force coefficient magnitude for G16. Although the initial normal force coefficient slope is accurately predicted at both Mach numbers, the nonlinear aspects of the curves are underpredicted for this fin configuration. This indicates the nonlinear formulation utilized in predicting normal force coefficient does not accurately model the effects of chord length increase on a fin. Axial force prediction results are very similar to those for G15. At Mach 0.5, the CA prediction is accurate throughout the angle of attack range, while at Mach 0.7, the coefficient is underpredicted.

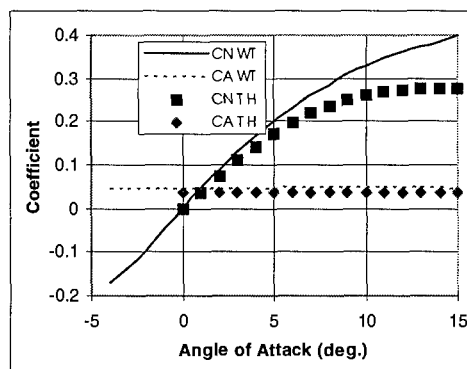


Figure 17. CN and CA Comparisons for Fin G16 at Mach 0.5.

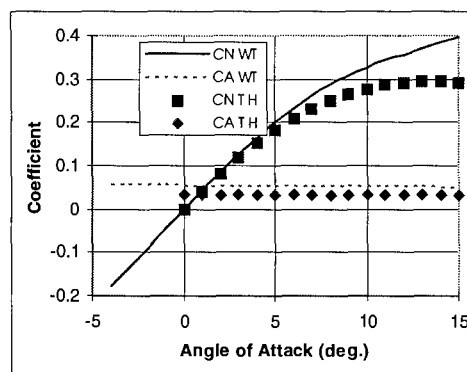


Figure 18. CN and CA Comparisons for Fin G16 at Mach 0.7.

Excellent prediction results were obtained for fin G12 at Mach 0.7. As indicated in Figure 19, both linear and nonlinear aspects of the normal force coefficient curve and the magnitude of the axial force coefficient were accurately predicted throughout the angle of attack range.

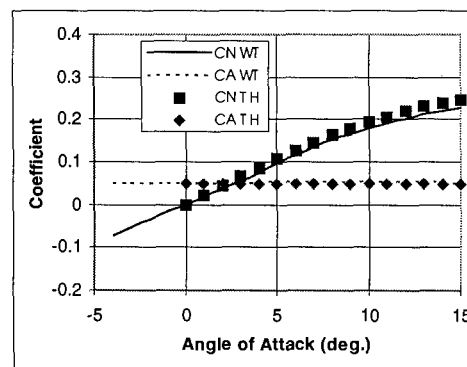


Figure 19. CN and CA Comparisons for Fin G12 at Mach 0.7.

Theoretical formulations developed for supersonic Mach numbers also provided reasonably accurate prediction results. Figures 20 through 23 illustrate the CN and CA comparisons for various grid fins in the horizontal position at Mach 2.5. As with the subsonic experimental data, these data were taken directly from the fin balance and therefore include the effects of body upwash.

As indicated in the various illustrations, grid fin configurations exhibit much more linear aerodynamic behavior in the supersonic Mach regime. As a result, inaccuracies in normal force coefficient prediction due to the

nonlinear semiempirical formulation are not nearly as significant as in the subsonic case. In all cases, the initial normal force coefficient slope is slightly overpredicted. However, discrepancies between the theoretical and experimental curves are less than 15% throughout the angle of attack range.

Axial force prediction using the supersonic formulations were consistently accurate for all grid fin configurations investigated, although the slight drop off in axial force magnitude with increasing angle of attack was not captured. This result occurs because the fin axial force was assumed independent of angle of attack in development of the formulation.

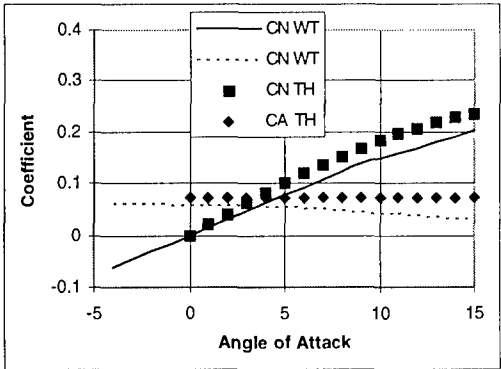


Figure 20. CN and CA comparisons for G12 at Mach 2.5.

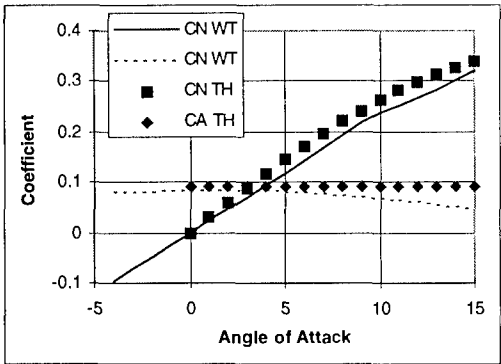


Figure 21. CN and CA comparisons for G13 at Mach 2.5.

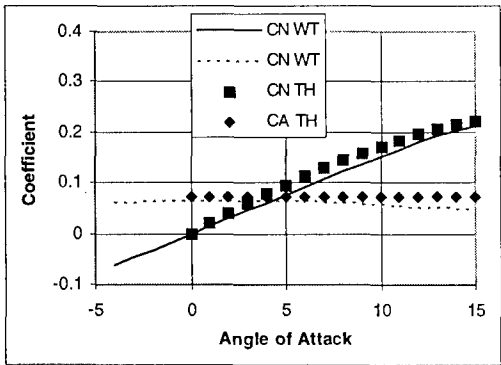


Figure 22. CN and CA comparisons for G15 at Mach 2.5.

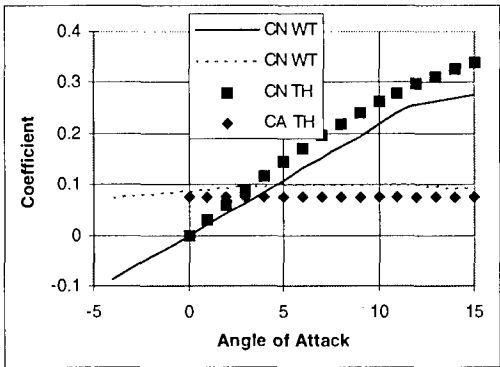


Figure 23. CN and CA comparisons for G16 at Mach 2.5.

Experimental data for fin orientations other than cruciform are somewhat limited, however, data were obtained for fin G13 in the "X" orientation at Mach 0.7 and 2.5. Prediction results for these fin orientations were very similar to those already presented. Figure 24 illustrates fin CN and CA coefficients for fin G13 rolled 45 deg. from the horizontal position. Similar to the cruciform orientation results, the linear aerodynamic portion of the curve is correctly predicted and nonlinear aspects of the curve are significantly underpredicted. Axial force coefficient is again predicted accurately throughout the angle of attack range.

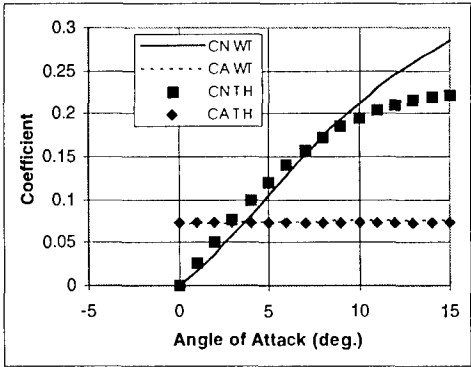


Figure 24. CN and CA comparisons for G13 at a 45 deg. Roll Angle at Mach 0.7.

Theoretical approximations at Mach 2.5 for fin G13 at a 45 deg. roll angle are presented with appropriate experimental data in Figure 25. As indicated, the normal force coefficient is accurately predicted throughout the angle of attack range, while the axial force is again overpredicted.

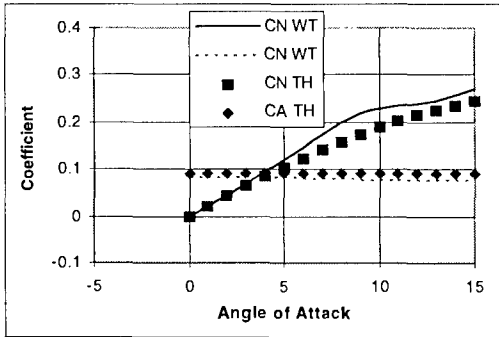


Figure 25. CN and CA comparisons for G13 at a 45 deg. Roll Angle at Mach 2.5.

6.0 CONCLUSIONS

1) The theoretical formulations developed for predicting the aerodynamic characteristics of grid fins in subsonic flow appear adequate for preliminary design purposes. Vortex lattice theory provides an adequate estimation of grid fin normal and axial force in the linear aerodynamic range provided body upwash terms are included. Empirical formulations developed to extend aerodynamic prediction capability to higher angles of attack provide good agreement with experimental data for a wide variety of grid fin configurations.

2) Theoretical formulations developed for application in the supersonic Mach regime are also adequate for preliminary grid fin design purposes. Application of a slightly modified version of Evvard's theory provides adequate estimations of grid fin lift curve slopes in the linear aerodynamic range. Empirical formulations developed to extend prediction capability to higher angles of attack provide good agreement with available experimental data.

3) Flowfield structure within the grid fin cells is not well understood in the transonic Mach regime, complicating the development of adequate prediction theories. However, initial theory development for transonic Mach numbers both above and below Mach 1.0 are promising.

4) Available experimental data have not been collected with theoretical development as a primary concern. As a result, separate validation of the various components effecting the force and moment characteristics of grid fins such as body upwash and fin carryover are not yet possible. Additionally, the influence of grid fin roll orientation on the force and moment characteristics is not well understood. Future experimental analysis and theoretical development which address these components separately should provide a more accurate means of predicting grid fin aerodynamic characteristics.

7.0 REFERENCES

1. Belotserkovsky, S. M., Odnovol, L.A., et.al, "Wings With Internal Framework", Machine Translation, FTD-ID(RS)T-1289-86, Foreign Technology Division, Feb. 1987.
2. Washington, W. D. and Booth, P. F., "Post Test Report for a Grid Fin Technology Wind Tunnel Test," Technical Report -RD-SS-89-6, US Army Missile Command, Redstone Arsenal, AL, July 1989.
3. Washington, W. D. and Miller, M. S., "Grid Fins - A New Concept for Missile Stability and Control," AIAA 93-0035, January 1993.
4. Burkhalter, J.E., Frank, Harris M., "Grid Fin Aerodynamics for Missile Applications in Subsonic Flow," Journal of Spacecraft and Rockets, Vol. 33, No. 1, Jan-Feb 1996.
5. Burkhalter, J.E., "Validation of Aerodynamic Prediction Methods for Grid Fins in Subsonic Flow," Contractor Final Report, Auburn University, U.S. Army Missile Command Contract No. DAAL03-91-C-0034, Jan., 1997.
6. Roskam, J., Methods for Estimating Drag Polars for Subsonic Airplanes, Published by the Author, 519 Boulder, Lawrence, Kansas, 1971.
7. Washington, W.D. and Booth, P. F., "Wind Tunnel Data Analysis for a Curved Grid Fin Concept," TTCP Technical Panel W-2, Internal Rept., Eglin AFB, FL, May 1990.
8. Miller, M.S., "An Experimental Investigation of Grid Fin Drag Reduction Techniques" AIAA 94-1914, June 1994.
9. Washington, W. D., Miller, M. S., "Experimental Investigation of Grid Fin Aerodynamics: A Synopsis of Nine Wind Tunnel and Three Flight Tests", AGARD Applied Vehicle Technology Panel Symposium and Meeting on Missile Aerodynamics, Paper 10, May 1998.
10. Burkhalter, J.E., "Characterization of Grid Fin Aerodynamics for Subsonic Flow," Final report, Cotntract No. DAAL03-91-C-0034, TCN Number:95-197, Scientific Services Program, AMSMI-RD-SS-AT, Redstone Arsenal, AL, Sept. 1995.
11. Burkhalter, J. E., "Grid Fins in Supersonic Flow," Contractor Final Report, Contract No. DAAH01-92-D-R002, DO Number NRC 0211, AMSMI-RD-SS-AT, Redstone Arsenal, AL, Sept. 1994.